

Clinical Risk Assessment Report: Patient 4

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 15.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.69x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 2.4%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.52 [Bottom 18%] (+1.9%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 1.06 [Top 14%] (+1.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -1.76 [Bottom 6%] (+1.4%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.26 [Avg (35th %ile)] (+1.1%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

3. Protective / Mitigating Factors (Reducing Risk)

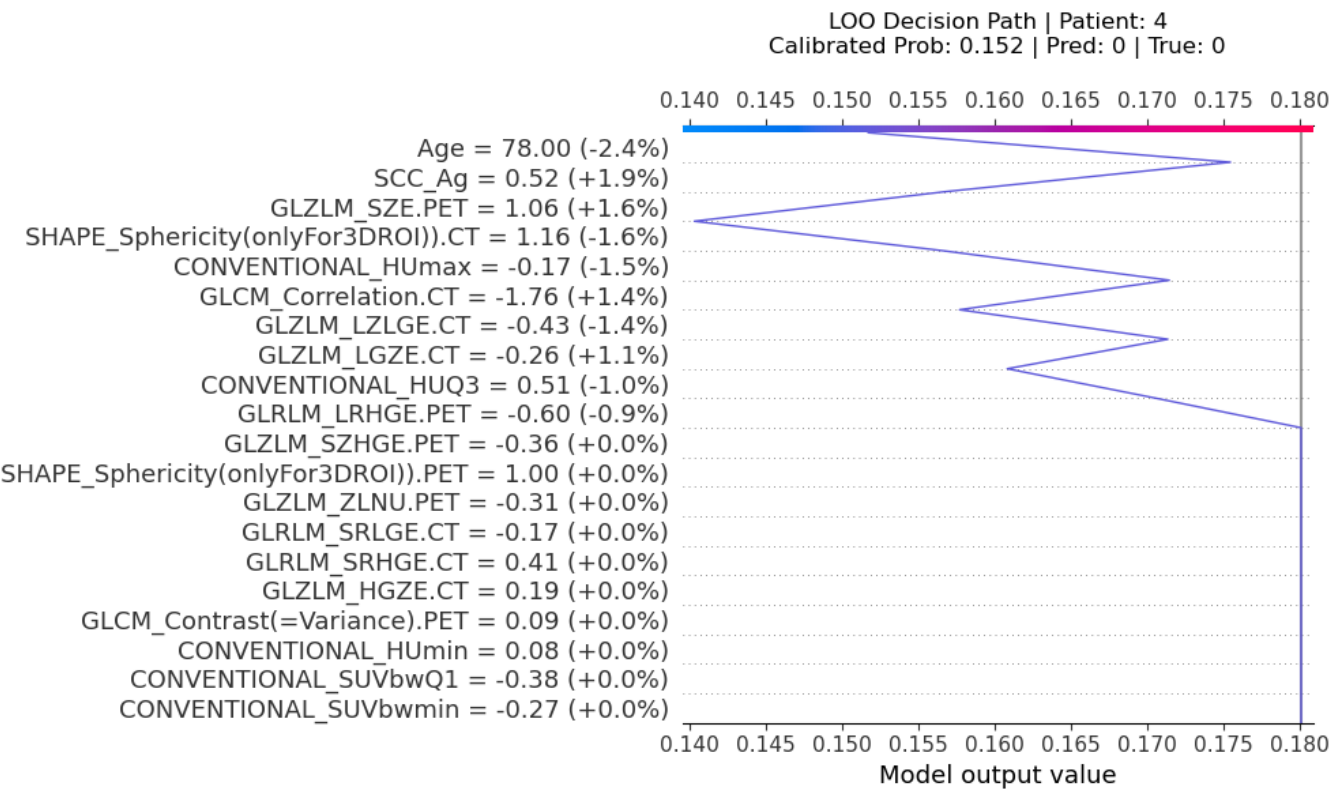
Patient Age (Age) **Value: 78.0 [Top 4%] (-2.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 1.16 [Top 13%] (-1.6%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.17 [Avg (38th %ile)] (-1.5%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.43 [Bottom <1%] (-1.4%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

